



ISLAMIC UNIVERSITY OF TECHNOLOGY

Electronics Mini Project

Team UNBIASED Transistor

Final Project Report

Implementation of a 1-to-4 Demultiplexer Using BJTs

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Project Objective and Application:

This project implements a **1-to-4 demultiplexer** built entirely from discrete **BC547 NPN transistors**, instead of using packaged logic ICs. The final circuit reaches its four outputs by cascading three smaller 1-to-2 demultiplexer stages, each built from transistor level inverters and NAND configured gates. The circuit takes a **single digital input** and routes it to **one of four output** lines depending on the state of two select lines, S_1 and S_0 . Building the logic from individual transistors rather than an integrated chip was done to show how basic digital logic functions, and small building block circuits, can be combined into a larger working system.

A BJT acts as a **switch** by toggling between **Cut-off** and **Saturation** to route logic signals. In contrast, it functions as an **amplifier** when biased in the **Active Region**, where a small input base current controls a larger collector current. In digital circuits like a demultiplexer, switching isolates or opens specific signal channels, while amplification buffers the routed signal to preserve voltage integrity across output loads.

A demultiplexer performs the reverse operation of a multiplexer: instead of combining several inputs into one output, it takes a single input and directs it to one of several outputs, with only one active output at a time. This function is useful anywhere a single data source needs to reach different destinations at different times, such as routing data in communication systems, selecting between memory blocks, or driving separate output devices from one shared control line.

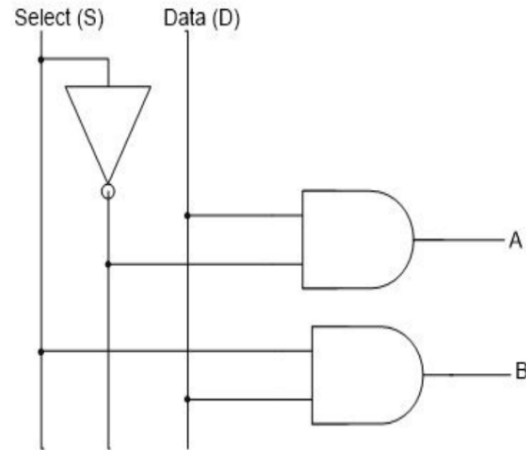
Circuit Diagram:

Two different transistor layouts were tried during development. The first attempt built the four output gates more directly off the input and select lines. Testing showed that cascading three smaller 1-to-2 demultiplexer stages instead, following a widely used BJT logic gate reference design built from

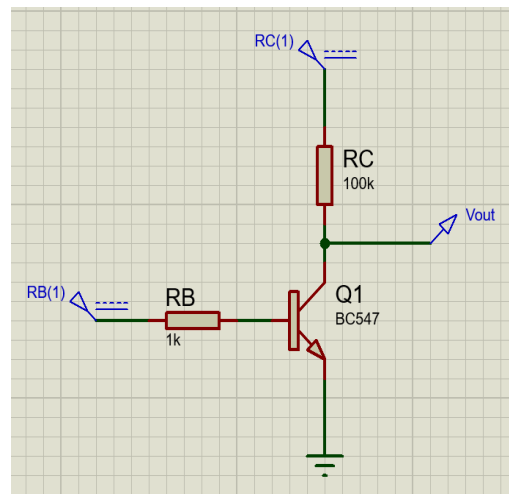
NAND configured transistor pairs, gave noticeably cleaner output, so the final build uses that cascaded arrangement.

First, we looked at the logical operation behind a 1-2 demux which consists of logical AND, NAND and NOT operations.

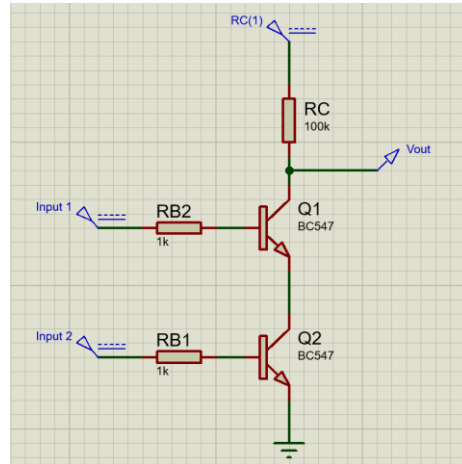
Logic diagram of 1-2 demux:



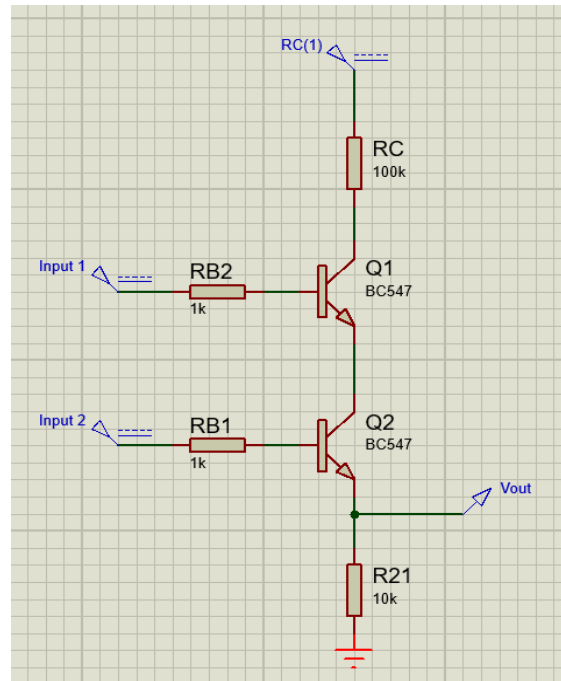
NOT gate using BJT:



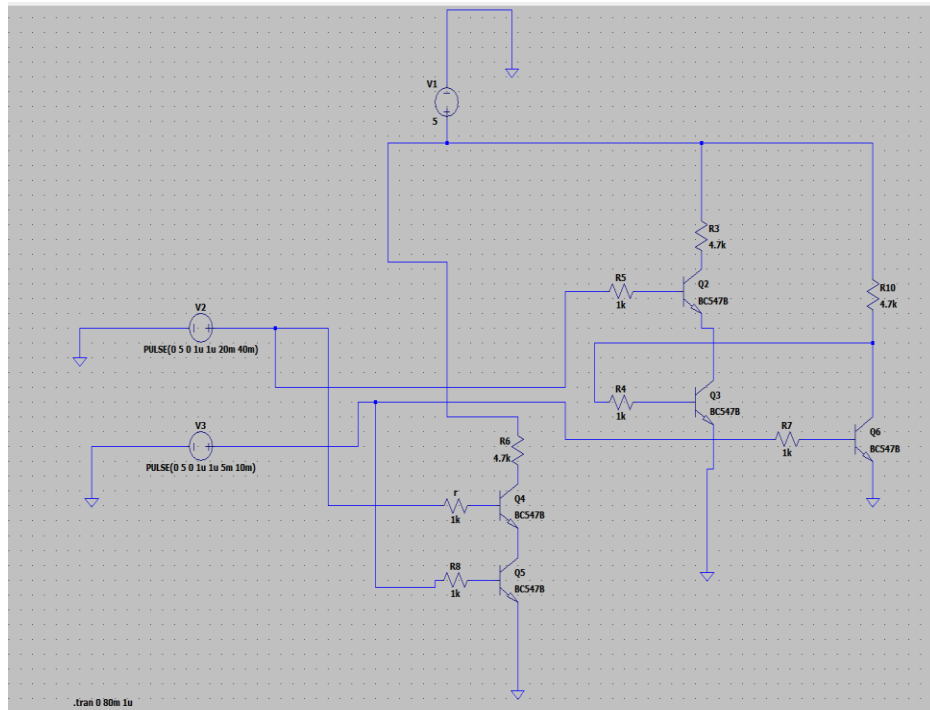
NAND gate using BJT:



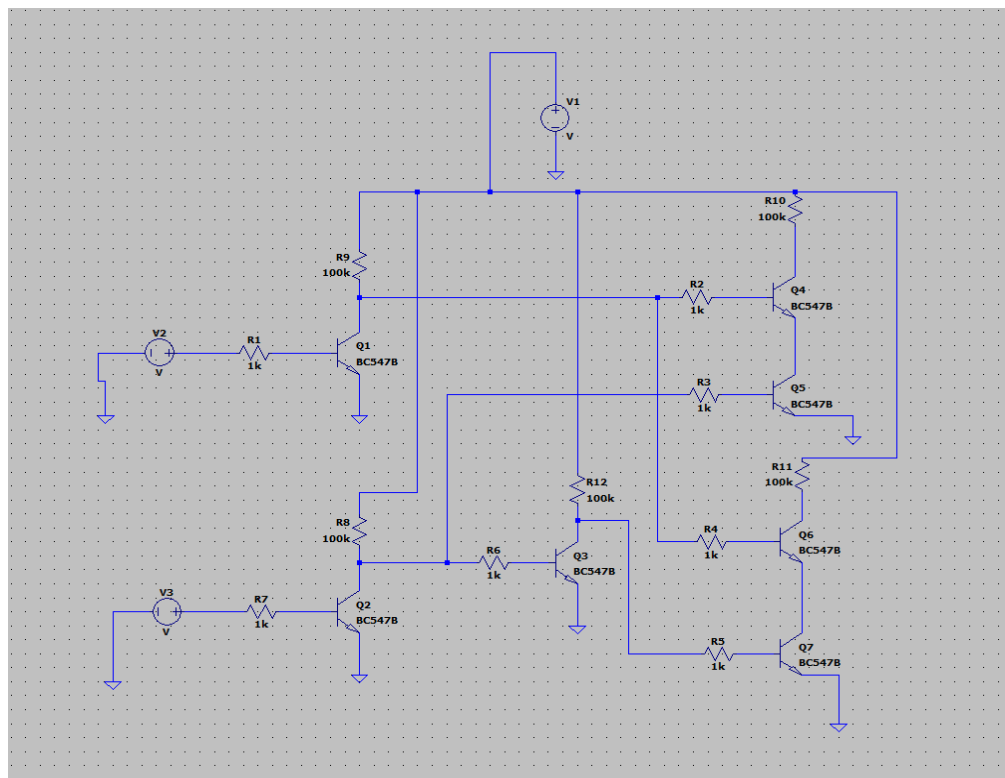
AND gate using BJT:



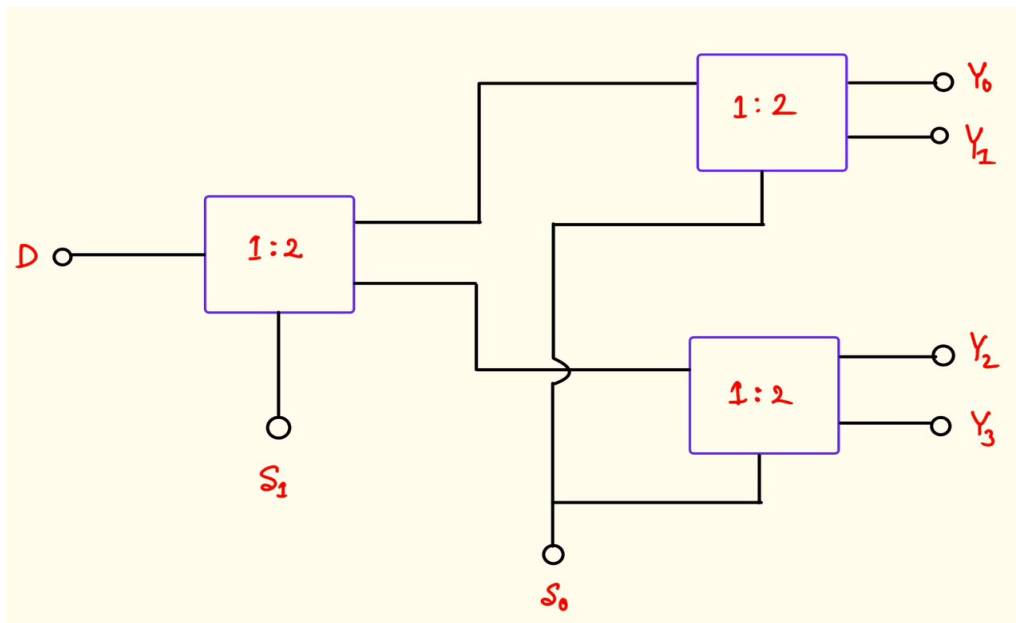
Using these configurations, we constructed the 1-to-2 demux following circuit diagram:



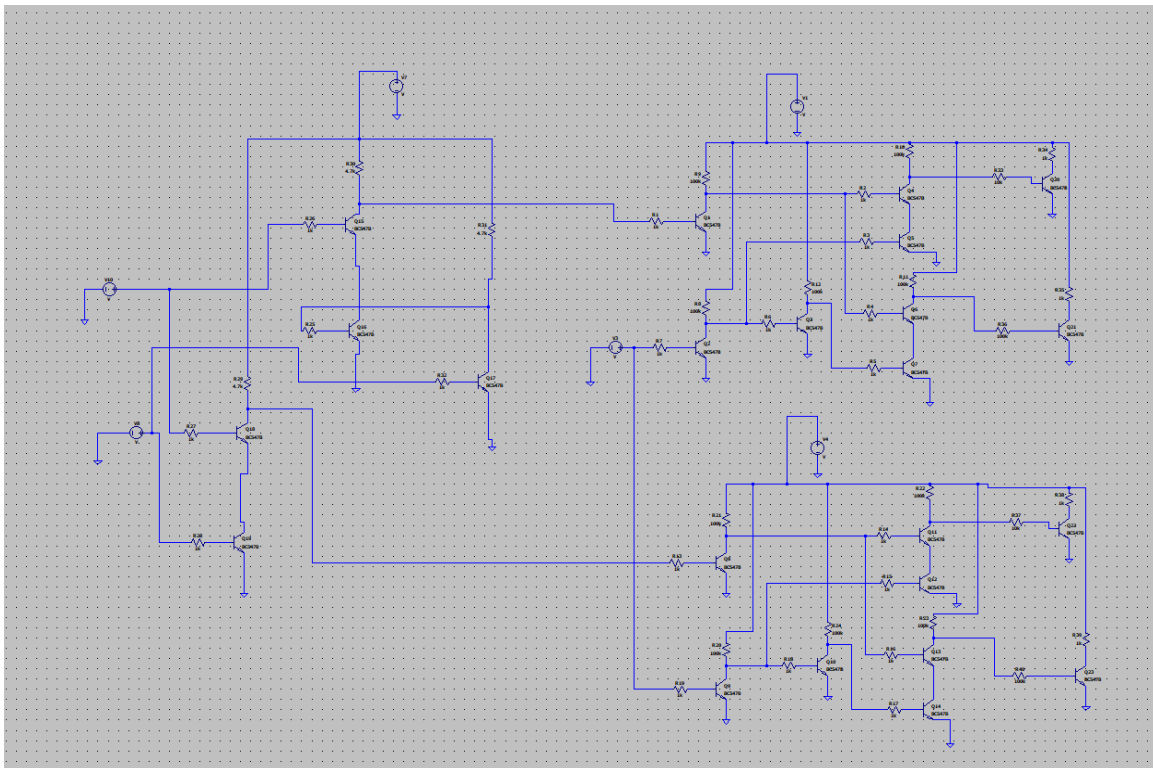
An alternative model for 1-2 demux:



Using the modular combination of the 1-2 demux we can construct the 1-4 demux as follows:



The schematic of 1 to 4 is given below:



Working Principle:

With two select lines, there are four possible combinations of S_1 and S_0 : 00, 01, 10, and 11. Each combination must activate exactly one output line and keep the remaining three low, following this mapping:

- $S_1 S_0 = 00$ activates Y_0
- $S_1 S_0 = 01$ activates Y_1
- $S_1 S_0 = 10$ activates Y_2
- $S_1 S_0 = 11$ activates Y_3

Rather than building four separate gates directly off the input and both select lines, the final circuit reaches this mapping by cascading three 1-to-2 demultiplexer stages. The first stage takes the data input D and splits it into two branches using S_1 , so one branch carries D forward when S_1 is low and the other carries D forward when S_1 is high. Each of those two branches then feeds a second stage split by S_0 : the branch selected by S_1 low produces Y_0 when S_0 is low and Y_1 when S_0 is high, and the branch selected by S_1 high produces Y_2 when S_0 is low and Y_3 when S_0 is high. In total the circuit uses three of these 1-to-2 stages, one at the first level and two more at the second level.

Each 1-to-2 stage is built from three BC547 transistors wired as simple inverters, using a base resistor from the input, a collector pull up resistor to $+V$, and a grounded emitter, plus four more BC547 transistors wired as two NAND configured pairs that combine the inverted and non-inverted select signal with the data line to produce the two stage outputs. Every transistor in the logic stages is operated purely between cutoff and saturation, so that it behaves as a clean switch rather than a linear amplifier at this point in the circuit. Altogether the three cascaded stages use twenty-one BC547 transistors for the core logic, plus four more BC547 transistors wired as common emitter amplifiers, one on

each of the four final outputs, to restore the signal strength lost across the two stages of switching between the input and any given output.

System Overview:

Input Section: Accepts a single digital input signal (D) and two select inputs (S₁, S₀), providing the control signals needed for output selection.

Demultiplexer Logic Section: Implements the demultiplexer as three cascaded 1-to-2 stages built from BC547 inverters and NAND configured transistor pairs, with S₁ controlling the first stage split and S₀ controlling both second stage splits, and includes a common emitter amplifier on each final output to restore signal strength lost through the two stages of switching.

Output Section: Produces one active output line corresponding to the selected address while the other three remain low, using LEDs to indicate which channel is active

Role of the BC547 Transistors:

- Acts as the switching element inside every inverter and NAND pair across all three cascaded stages.
- Builds the three inverters and two NAND configured pairs that make up each 1-to-2 stage.
- Selects which output branch is active, first by S₁ at the top stage and then by S₀ at the second stage.
- Provides the signal inversion needed to realize the NAND based routing at each stage.

- A separate group of four BC547 transistors, wired as common emitter amplifiers, restores the signal on each final output line after it has passed through two stages of switching.
- The output transistors sink current to drive the LEDs safely.

Component List & Specifications:

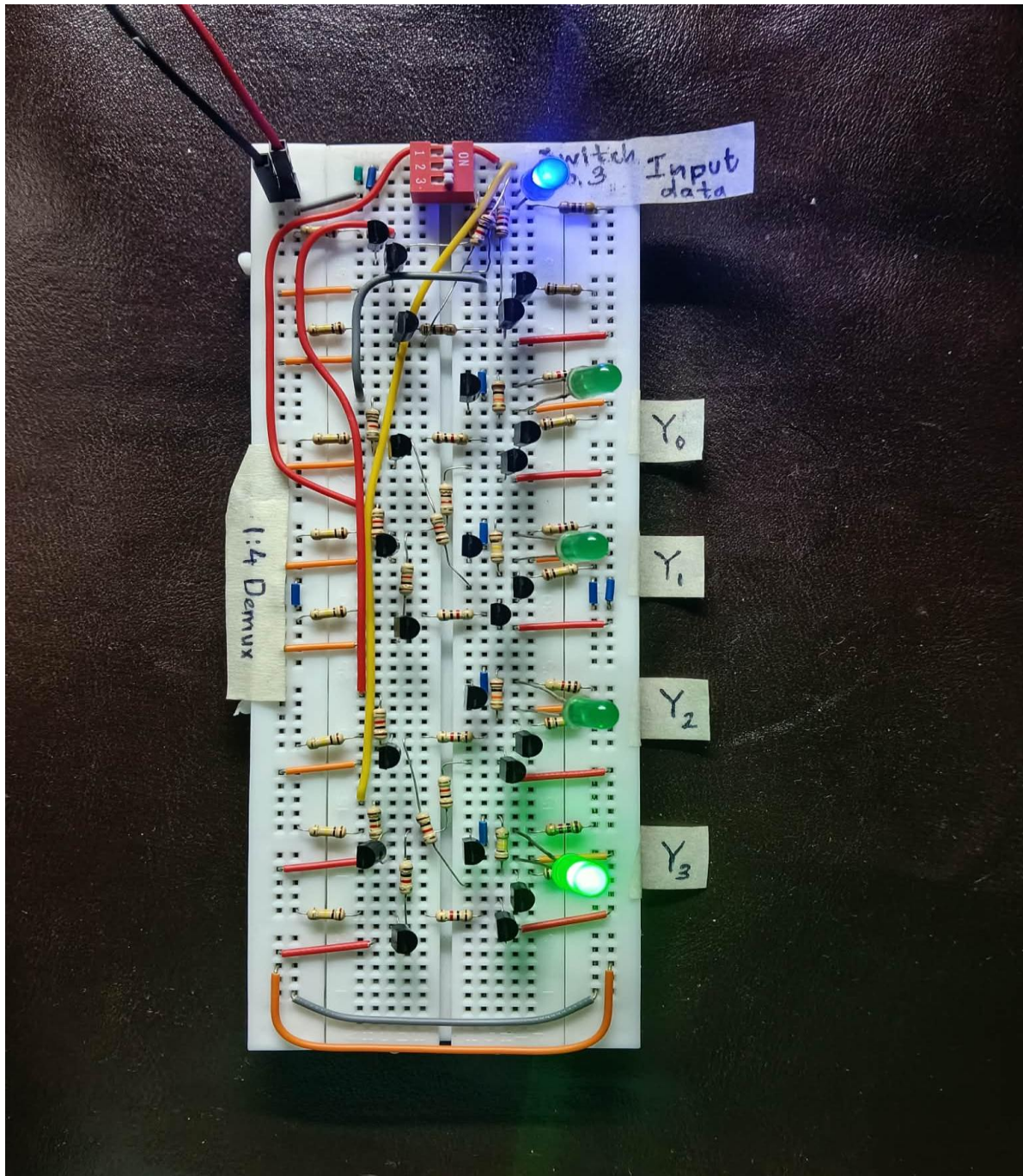
Sl. No.	Component	Specification	Qty.	Purpose
1	BC547 NPN Transistor	BC547B	~25	Builds the inverters and NAND pairs in the three cascaded 1-to-2 stages, plus one common emitter amplifier per output
2	Resistors	100k Ω , 4.7k Ω , 1k Ω	~40	Collector pull ups and base bias for the logic stages, and biasing for the amplifier stage
3	LEDs	5mm	5	Four green LEDs for the Y0 to Y3 outputs, plus one LED used as an input status indicator
4	DC power supply	+5 V	1	Powers the circuit
5	Breadboard	Standard	1	Circuit prototyping
6	Jumper wires	Male to male	Many	Electrical connections
7	DIP Switch	3 position	1	Provides the D, S ₁ , and S ₀ input signals
8	Multimeter	Digital	1	Measuring output voltage and signal loss at each stage
9	Oscilloscope	Digital	1	Observing signal waveforms during testing

Design Considerations:

- The final circuit cascades three 1-to-2 demultiplexer stages instead of building the four output gates directly, since testing showed the cascaded layout gave cleaner output.
- All logic gates, the inverters and the NAND configured pairs, were built from discrete BC547 transistors rather than integrated logic ICs, to work directly with transistor level switching.
- The circuit runs from a +5 V DC supply, matching standard digital logic levels.
- BC547 transistors in the logic stages were operated between cutoff and saturation to act as clean electronic switches.
- A common emitter amplifier stage was added on each final output to compensate for the signal loss measured across the two cascaded stages.
- Some resistor values were changed from the original reference design, for example from 100 k Ω to 4.7 k Ω in places, to support the amplifier stage.
- LEDs were placed at each output to give a visual confirmation of the active channel during testing.
- The design assumes binary input throughout and is not intended to route analog signals.

Project Pictures:

Completed circuit in breadboard:



Experimental Results and Observations:

S ₁	S ₀	D	Y ₀	Y ₁	Y ₂	Y ₃
0	0	1	1	0	0	0
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	1
X	X	0	0	0	0	0

The circuit correctly routed the input to the intended output for every combination of S₁ and S₀, matching the expected logic behavior shown above. Several practical issues showed up once the gates were tested at the transistor level, which the truth table alone does not capture.

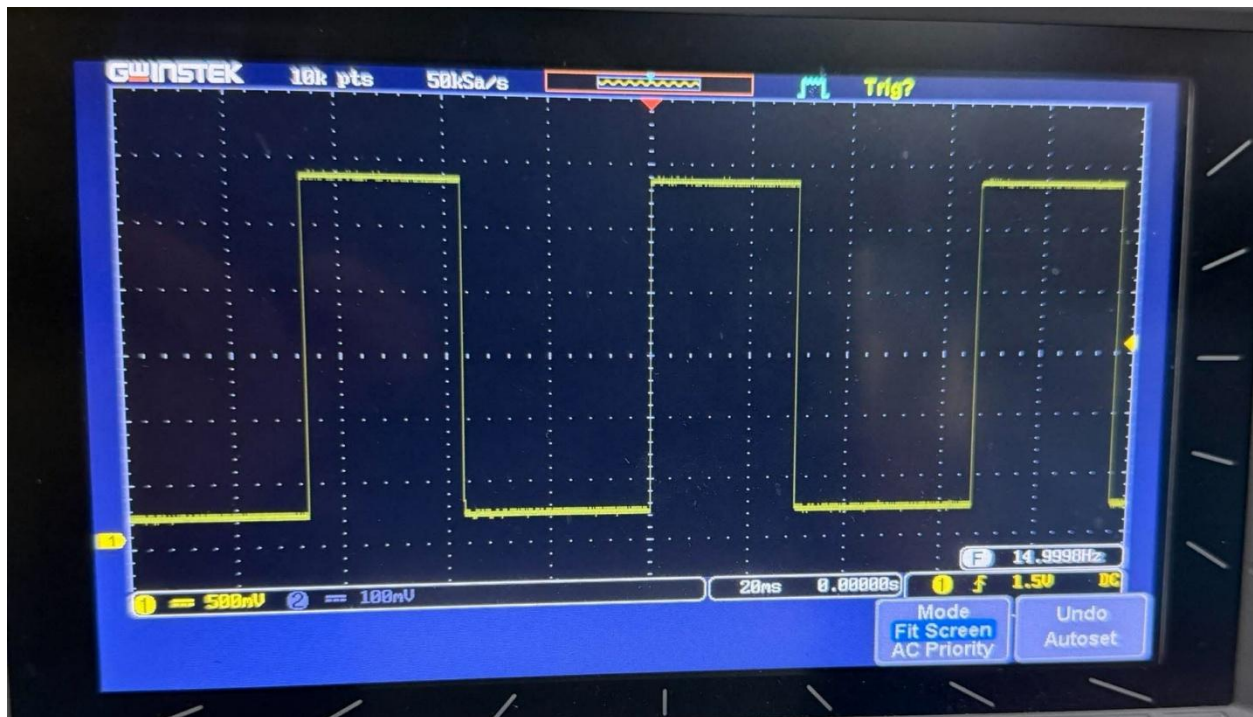
The signal weakened noticeably as it passed through the two cascaded stages between the input and any given output. This came from the collector emitter saturation voltage of each BJT stage, combined with the internal resistance of the components in the signal path, and the loss added up across both stages the signal passed through before reaching an output. Because of this drop, we added a common emitter amplifier at each of the four output terminals to restore the signal before it reached the LED.

Even with the amplifiers in place, the four outputs did not behave identically. Voltage measurements taken with a multimeter showed a different amount of loss on each output line. Since every output passes through the same number of cascaded stages regardless of which line is selected, this unevenness most likely came from small variations between individual transistors and resistors across the three separate stage instances, rather than from any one channel having a longer signal path. Getting fully consistent output across all four channels proved difficult, and amplification only partly closed that gap. This was visible directly on the finished board, where the four output LEDs did not all reach the same brightness for an equivalent logic state.

We also found that one or more of the LEDs used in the build had a manufacturing defect that caused visible brightness variation on its own, separate from the actual output signal level. This made it harder to judge, from brightness alone, whether a dim output was caused by the circuit or by the LED.

Testing with square

To check the circuit under a continuously repeating signal rather than a simple switch toggle, we also drove the input with a function generator while watching the result on an oscilloscope, using both a square wave and a sine wave.



- Square wave input at approximately 15 Hz, measured directly on the oscilloscope as 14.9998 Hz, and the function generator amplitude set to approximately 3 V.

With the square wave input, the oscilloscope showed a clean, consistent pulse train on channel 1, confirming that the circuit reproduced a digital square wave correctly at this frequency.

Partway through the build, we also found that cascading three 1-to-2 demultiplexer stages, based on the reference design described in the circuit diagram section, gave cleaner output than our original single stage gate layout, and the final version reflects that change.

Challenges Encountered and Improvements That We Made:

Challenges

- Signal loss through the cascaded transistor stages, caused by collector emitter saturation voltage and internal component resistance building up across two stages of switching.
- Uneven voltage loss between the four output channels, most likely from component-to-component variation across the three separate stage instances.
- Brightness variation in the LEDs caused by a defect in the LEDs themselves rather than the circuit.
- The original single stage gate layout was not optimal and had to be redesigned into the cascaded 1-to-2 stage arrangement partway through the build.
- While observing by function generator input, we saw small ripple/noise in our output waveform.

Possible Improvements

- Using matched transistors, or resistors with tighter tolerance, across the three stage instances could help even out the channel-to-channel voltage differences.
- To adjust the voltage drop, we added a common emitter amplifier to each of the output terminals.

- Replacing the defective LEDs would remove that source of measurement error from future testing.
- Testing each of the three stages individually before final assembly could catch layout or component issues earlier, instead of discovering them only after the full circuit was built
- We can add a capacitor to the power rails to remove the noise completely.